

Introduction to Machine Learning

Christos Dimitrakakis

September 15, 2026

Outline

The problems of Machine Learning (1 week)

- Introduction

Estimation

- Answering a scientific problem

- Pandas and dataframes

- Additional notes on pandas

- Single variable models

- Two variable models

Statistics, validation and model selection

Course summary

- Course Contents

For next week

- Assignment

- Reading

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Pandas and dataframes

Additional notes on pandas

Single variable models

Two variable models

Statistics, validation and model selection

Course summary

Course Contents

For next week

Assignment

Reading

Machine Learning And Data Mining

The nuts and bolts

- ▶ Models
- ▶ Algorithms
- ▶ Theory
- ▶ Practice

Workflow

- ▶ Scientific question
- ▶ Formalisation of the problem
- ▶ Data collection
- ▶ Analysis and model selection

Types of statistics / machine learning problems

- ▶ Classification
- ▶ Regression
- ▶ Density estimation

Machine learning

Data Collection

- ▶ Downloading a clean dataset from a **repository**
- ▶ **Scraping** data from the web
- ▶ Conducting a **survey**
- ▶ Performing **experiments**, and obtaining measurements.

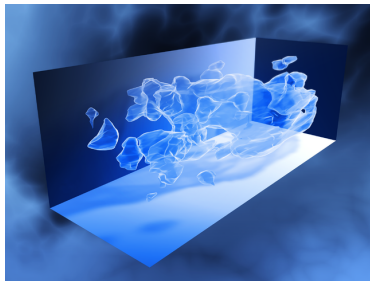
Modelling

- ▶ Simple: the bias of a coin
- ▶ Complex: a language model.
- ▶ The model depends on the data and the problem

Algorithms and Decision Making

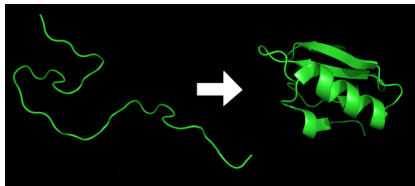
- ▶ We want to use models to make decisions.
- ▶ Decisions are made every step of the way.
- ▶ Both humans and algorithms can make decisions.

The main problems in machine learning and statistics



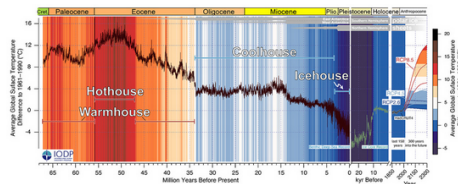
Matter

Dark

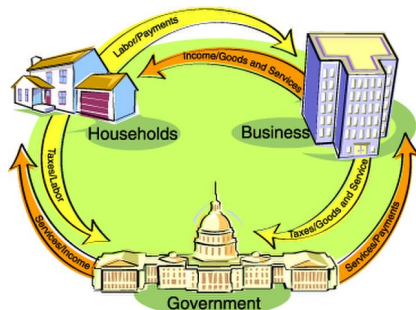


Protein Folding

Christos Dimitrakakis



Climate Modelling



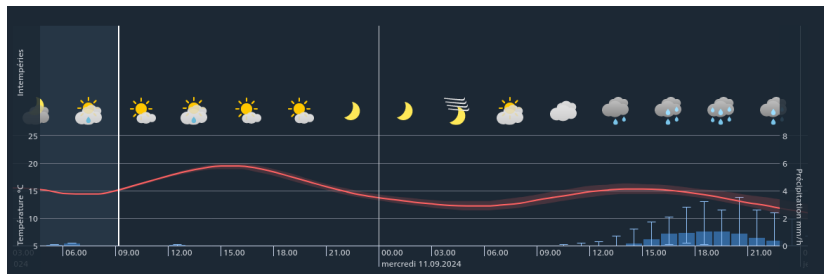
Economic Policy

Introduction to Machine Learning

September 13, 2026

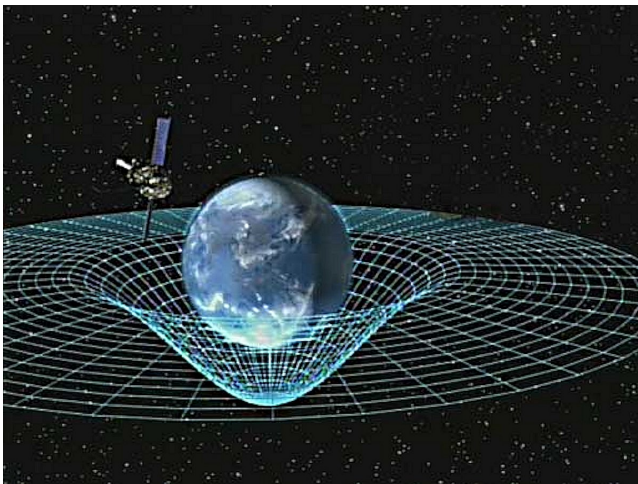
6 / 49

Prediction



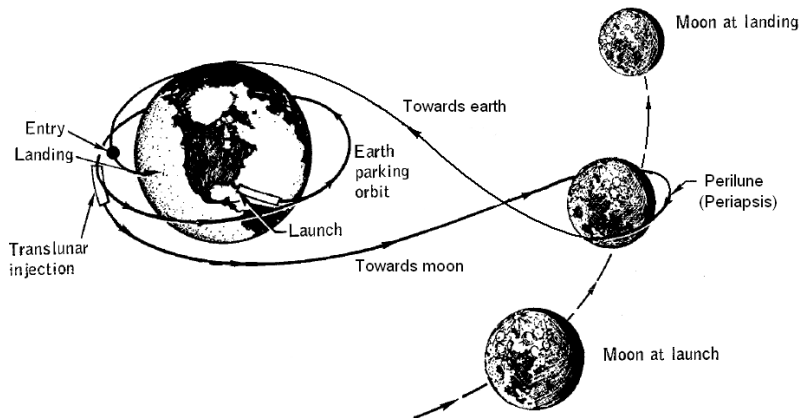
- ▶ Will it rain tomorrow?
- ▶ How much will bitcoin be worth next year?
- ▶ When is the next solar eclipse?

Inference



- ▶ What is the law of gravitation?
- ▶ Where is the spaceship now?
- ▶ Does my poker opponent have two aces?

Decision Making



- ▶ What data should I collect?
- ▶ Which model should I use?
- ▶ Should I fold, call, or raise in my poker game?
- ▶ How can I get a spaceship to the moon and back?

./fig/artemis.gif

The need to learn from data

Problem definition

- ▶ What problem do we need to solve?
- ▶ How can we formalise it?
- ▶ What properties of the problem can we learn from data?

Data collection

- ▶ **Why** do we need data?
- ▶ **What** data do we need?
- ▶ How **much** data do we want?
- ▶ **How** will we collect the data?

Modelling and decision making

- ▶ How will we **compute** something useful?
- ▶ How can we use the model to make **decisions**?

Course Material

Moodle

- ▶ Assignments and proejct
- ▶ Additional reading material
- ▶ Asking questions

Course Github

- ▶ .org files for notes, PDF for slides
- ▶ source code for examples



Course literature

- ▶ An Introduction to Statistical Learning with Python
- ▶ Book chapters will be mentioned in the course



Assignments, teaching and questions

Course structure

- ▶ In-class exercises.
- ▶ Take-home assignments (ungraded).
- ▶ Quizzes.
- ▶ In-class group work.

Other questions

- ▶ Use Moodle for technical/administrative questions: That way everybody gets the same information.
- ▶ Use email for personal problems or extra help, if the moodle is not enough.
- ▶ Complicated questions can be answered at the next lecture.

Office hours

- ▶ Tuesdays 13:15-14:00: book with an email to avoid clashes.
- ▶ Email me for an appointment outside those hours.

The project pipeline

- ▶ Builds on in-class exercises and take-home assignment
- ▶ Every group must create a github repository and commit regularly.
- ▶ A short presentation will be given.

Key points

- ▶ Define a scientific question.
- ▶ Identify relevant variables and their relationships.
- ▶ Generate data from counterfactual simulations.
- ▶ Analyse the data from the different counterfactuals.

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Pandas and dataframes

Additional notes on pandas

Single variable models

Two variable models

Statistics, validation and model selection

Course summary

Course Contents

For next week

Assignment

Reading

Problem definition

- ▶ Example: Health, weight and height

Example (Health questions regarding height and weight)

- ▶ What is a normal height and weight?
- ▶ How are they related to health?
- ▶ What variables affect height and weight?

Define a research question

Find a **non-sensitive** variable that we can easily measure via a survey, e.g. related to sleep, smoking, exercise, food, politics, sports, hobbies etc.

- ▶ Discuss in small groups and post suggestions
- ▶ We then vote for what to measure

Data collection

Think about **which variables** we need to collect to answer our **research question**.

Necessary variables

The variables we need to know about

- ▶ Weight
- ▶ Height
- ▶ Dependent: (health/vote/opinion/salary)

Auxiliary variables

Measurable factors related to the variables of interest

Possible confounders

+Hidden factors that might affect variables. We will formalise these dependencies with **graphical models** later in the course — for now, just try to list them.

Class data and variables

- ▶ The class enters their data into the excel file.



- ▶ Pay attention to the variables we wish to measure

Privacy

- ▶ Is the use of a pseudonym sufficient to hide your identity?

Variables

The class data looks like this

First Name	Gender	Height	Weight	Age	Nationality	Smoking
Lee	M	170	80	20	Chinese	10
Fatemeh	F	150	65	25	Turkey	0
Ali	Male	174	82	19	Turkish	0
Joan	N	5'11	180	21	American	4

- $\mathbf{X}$: Everybody's data
- x_t : The t -th person's data
- $x_{t,k}$: The k -th feature of the t -th person.
- x_k : Everybody's k -th feature
- Anything strange with this data?

Inconsistent data

You can see that the entries are inconsistent. In some cases we have Male, and others M. The value N maybe a typo, or may indicate Neutral? The height and weight seem to be in cm and kg for most people, but Joan seems to use feet'inches and pounds.

Raw versus neat data

- Neat data: $x_i \in \mathbb{R}^n$

Types of learning problems

Unsupervised learning (unconditional estimation)

- ▶ Predict the **gender** of an unknown individual.
- ▶ Predict the **height**.
- ▶ Predict the **height and weight**?

Types of learning problems

Unsupervised learning (unconditional estimation)

- ▶ Predict the **gender** of an unknown individual.
- ▶ Predict the **height**.
- ▶ Predict the **height and weight**?

Supervised learning problems (conditional estimation)

- ▶ Classification: Can we predict gender from height/weight?
- ▶ Regression: Can we predict weight from height and gender?
- ▶ In both cases we predict **output** variables from **input** variables

Types of learning problems

Unsupervised learning (unconditional estimation)

- ▶ Predict the **gender** of an unknown individual.
- ▶ Predict the **height**.
- ▶ Predict the **height and weight**?

Supervised learning problems (conditional estimation)

- ▶ Classification: Can we predict gender from height/weight?
- ▶ Regression: Can we predict weight from height and gender?
- ▶ In both cases we predict **output** variables from **input** variables

Variables

- ▶ **Input** variables: aka features, predictors, independent variables
- ▶ **Output** variables: aka response, dependent variables, labels, or targets.
- ▶ The input/output dichotomy only exists in **some prediction problems**.

Looking at the class data

Reading the class data

```
import pandas as pd
X = pd.read_excel("data/class.xlsx")
X["Height"]      # bx_k : the Height feature, for everybody
X.loc[2]         # x_t  : individual 2, all features
```

- ▶ A DataFrame is **not the same** as a numpy array.
- ▶ Columns keep their names ("Height")
- ▶ Rows can be accessed through index.

Summarising the class data

```
X.hist()
import matplotlib.pyplot as plt
plt.show()
```

- ▶ Check the histograms for inconsistent raw values: does anything look wrong?

Pandas and DataFrames

- ▶ Data in pandas is stored in a **DataFrame**
- ▶ DataFrame is **not the same** as a numpy array.

Core libraries

```
import pandas as pd
import numpy as np
```

Series: A sequence of values

```
# From numpy array:
s = pd.Series(np.random.randn(3), index=["a", "b", "c"])
# From dict:
d = {"a": 1, "b": 0, "c": 2}
s = pd.Series(d)
# accessing elements
s.iloc[2] #element 2
s.iloc[1:2] #elements 1,2
s.array # gets the array object
s.to_numpy() # gets the underlying numpy array
```

DataFrames

Constructing from a numpy array

```
data = np.random.uniform(size = [3,2])
df = pd.DataFrame(data, index=["John", "Ali", "Sumi"],
    columns=["X1", "X2"])
```

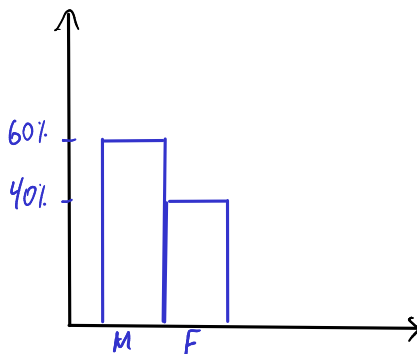
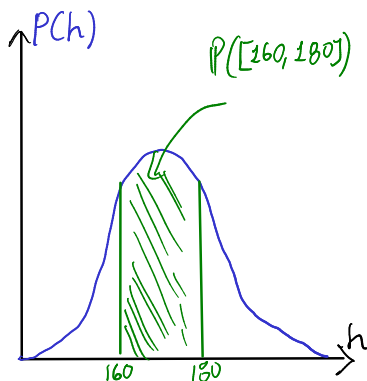
Constructing from a dictionary

```
d = { "one": pd.Series([1, 2], index=["a", "b"]),
      "two": pd.Series([1, 2, 3], index=["a", "b", "c"])}
df = pd.DataFrame(d)
```

Access

```
X["First Name"] # get a column
X.loc[2] # get a row
X.at[2, "First Name"] # row 2, column 'first name'
X.loc[2].at["First Name"] # row 2, element 'first name' of the series
X.iat[2,0] # row 2, column 0
```

Modelling single variables

Figure: $x \in \mathbb{N}$ Figure: $x \in \mathbb{R}$

Means using python

Example (Calculating the mean of our class data)

```
X.mean()           # mean of every column of the class DataFrame
X["Height"].mean() # just the height column
np.mean(X["Weight"]) # equivalent, via numpy
```

- ▶ The mean here is **fixed** because we calculate it on the same data.
- ▶ If we were to **collect new data** then the answer would be different.

Example (Calculating the mean of a random variable)

```
import numpy as np
y = np.random.gamma(170, 1, size=20) # 20 fresh samples -- not the
y.mean()
np.mean(y)
```

- ▶ The mean is **random**, so we get a different answer every time we re-run this.

Probabilities: Basic notions.

Probability space

- ▶ Defined on subsets of Ω , a random **outcome space**.
- ▶ Subsets A of Ω , $A \subseteq \Omega$, are called **events**.

Probability axioms

A probability P , is a function from subsets¹ of Ω to $[0, 1]$ and it functions like an area:

- ▶ $P(\Omega) = 1$ (whatever happens must be in Ω).
- ▶ $P(A) \geq 0$ is the **probability** of $A \subseteq \Omega$.
- ▶ If A, B are **mutually exclusive** ($A \cap B = \emptyset$), then:

$$P(A \cup B) = P(A) + P(B).$$

¹Formally, an algebra

Meaning of probability

Probabilities are usually subjective (exception: quantum physics).

Subjective probability rule

If we believe an event A is **more likely** than B then $P(A) > P(B)$.

Properties that follow.

- ▶ Implication: If A always happens if B happens, then every rational person should believe A is at least as likely as B .
- ▶ Transitive property: If A is more likely than B and B is more likely than C then A is more likely than C .

Special cases

The outcomes or universe Ω depends on what we are interested in, and how we generate the random outcomes.

Discrete Ω

- ▶ Consider throwing two coins. Then possible outcomes are $\Omega = \{HH, HT, TH, TT\}$ for heads or tails for each coin.
- ▶ Consider throwing one die. Then possible outcomes are $\Omega = [6] = \{1, 2, 3, 4, 5, 6\}$.
- ▶ Consider selecting an individual in Switzerland by randomly generating an AVS number: $\Omega = [10^{12}]$.

Continuous Ω

Consider throwing a die inside a 1×1 box. Then possible outcomes are in $\Omega = [0, 1]^2 \times [0, 360]^3$, because we need:

- ▶ 2 $[0, 1]$ variables for the location
- ▶ 3 $[0, 360]$ variables for the angle of rotation

Which is the most natural outcome space for throwing a die?

Random variables

Even though the most natural outcome space for a die throw is $[6]$, we actually observe the die as a 3D object in $[0, 1]^2 \times [0, 360]^3$. So we **map** the outcome in Ω onto $[6]$ if we want to play dice, i.e. we construct a **random variable (RV)** $x : [0, 1]^2 \times [0, 360]^3 \rightarrow [6]$.

Dice game

- ▶ Here consider $\Omega = [6]$ and $P(\omega) = 1/6$ for every outcome $\omega \in \Omega$.
- ▶ Define the RV $x : \Omega \rightarrow \{0, 1\}$ as

$$x(\omega) = \begin{cases} 1, & \text{if } \omega \geq 5 \\ 0, & \text{if } \omega < 5 \end{cases}$$

where 1 means winning, and 0 means losing.

- ▶ What is the probability that $x = 1$?

Probabilities of RVs

For any $x : \Omega \rightarrow X$, we can introduce a new probability measure P_x so that, for any set of outcomes $S \subseteq X$ of the **random variable**, we have:

$$P_x(S) = P(\{\omega \in \Omega : x(\omega) \in S\})$$

Introduction to Machine Learning

└ Estimation

└ Single variable models

└ Random variables

Random variables

Even though the most natural outcome space for a die throw is $[6]$, we actually observe the die as a 3D object in $[0, 1]^3 \times [0, 360]^\circ$. So we map the outcome in Ω onto $[6]$ if we want to play dice, i.e. we construct a **random variable** (RV) $x : [0, 1]^3 \times [0, 360]^\circ \rightarrow [6]$.

Dice game

- Here consider $\Omega = [6]$ and $P(\omega) = 1/6$ for every outcome $\omega \in \Omega$.
- Define the RV $x : \Omega \rightarrow (0, 1]$ as

$$x(\omega) = \begin{cases} 1, & \text{if } \omega \geq 5 \\ 0, & \text{if } \omega < 5 \end{cases}$$

- where 1 means winning, and 0 means losing.
- What is the probability that $x = 1$?

Probabilities of RVs

For any $x : \Omega \rightarrow X$, we can introduce a new probability measure P_x so that, for any set of outcomes $S \subseteq X$ of the **random variable**, we have:

$$P_x(S) = P(\{\omega \in \Omega : x(\omega) \in S\})$$

Alternative definition

- An equivalent definition: $x(\omega) = \mathbb{I}\{\omega \geq 5\}$,

uses the indicator function $\mathbb{I}\{\cdot\}$ which takes the value 1 when its argument is true and 0 otherwise. Probability of $x = 1$ Since we are only interested in $x = 1$, then $S = \{1\}$. As a **shorthand**, we usually write $P_x(1)$ instead of $P_x(\{1\})$.

$$P_x(1) = P(\{x(\omega) = 1 : \omega \in \Omega\})$$

Here, $\{x(\omega) = 1 : \omega \in \Omega\}$ is the set of values $\omega \in \Omega$ for which $x(\omega) = 1$. These are $\{5, 6\}$ in our example. Consequently,

$$P_x(1) = P(\{5, 6\}) = P(5) + P(6) = 1/6 + 1/6 = 1/3.$$

This second equality follows from the third axiom of probability, because $\{5\}$ and $\{6\}$ are mutually exclusive.

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$
- ▶ $\mathbb{E}_P[x]$: expectation of x under P (is the same for all t)

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$
- ▶ $\mathbb{E}_P[x]$: expectation of x under P (is the same for all t)

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$
- ▶ $\mathbb{E}_P[x]$: expectation of x under P (is the same for all t)

$$\mathbb{E}_P[x] = \sum_{\omega \in \Omega} x(\omega)P(\omega)$$

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$
- ▶ $\mathbb{E}_P[x]$: expectation of x under P (is the same for all t)

$$\mathbb{E}_P[x] = \sum_{\omega \in \Omega} x(\omega)P(\omega)$$

Definition (The sample mean)

- ▶ **i.i.d.** variables $x_1, \dots, x_t, \dots, x_T$: with $x_t = x(\omega_t)$, $\omega_t \sim P$.

One variable: expectations

Definition (The expected value)

- ▶ Ω : random outcome space
- ▶ P : distribution of outcomes $\omega \in \Omega$
- ▶ Random variable $x : \Omega \rightarrow \mathbb{R}$, and $\omega \sim P$
- ▶ $\mathbb{E}_P[x]$: expectation of x under P (is the same for all t)

$$\mathbb{E}_P[x] = \sum_{\omega \in \Omega} x(\omega)P(\omega)$$

Definition (The sample mean)

- ▶ **i.i.d.** variables $x_1, \dots, x_t, \dots, x_T$: with $x_t = x(\omega_t)$, $\omega_t \sim P$.
- ▶ The sample mean of $x_1, \dots, x_T$ is

$$\frac{1}{T} \sum_{t=1}^T x_t$$

The sample mean is $O(1/\sqrt{T})$ -close to $\mathbb{E}_P[x_t]$ with high probability.

Reminder: expectations of random variables

A gambling game

- ▶ [a] With probability 1%, you win 100 CHF
- ▶ [b] With probability 40%, you win 20 CHF.
- ▶ [c] Otherwise, you win nothing

Exercises

- ▶ What are the expected winnings if you play this game? (on paper)
- ▶ Implement a method in python that generates draws from this random variable, using a uniformly random outcomes in $[0, 1]$ as an input.

Introduction to Machine Learning

└ Estimation

└ Single variable models

└ Reminder: expectations of random variables

Reminder: expectations of random variables

A gambling game

- ▶ [a] With probability 1%, you win 100 CHF
- ▶ [b] With probability 40%, you win 20 CHF
- ▶ [c] Otherwise, you win nothing

Exercises

- ▶ What are the expected winnings if you play this game? (on paper)
- ▶ Implement a method in python that generates draws from this random variable, using a uniformly random outcomes in $[0, 1]$ as an input.

Solution

Calculation

- Let x be the amount won, then $x(a) = 100, x(b) = 20, x(c) = 0$.
- We need to calculate

$$\mathbb{E}_P(x) = \sum_{\omega \in \{a, b, c\}} x(\omega)P(\omega) = x(a)P(a) + x(b)P(b) + x(c)P(c)$$

- $P(c) = 59\%$, as $P(\Omega) = 1$. Substituting,

$$\mathbb{E}_P(x) = 1 + 8 + 0 = 9.$$

Models

Models as summaries

- ▶ They summarise what we can see in the data
- ▶ The ultimate model of the data **is** the data

Models

Models as summaries

- ▶ They summarise what we can see in the data
- ▶ The ultimate model of the data **is** the data

Models as predictors

- ▶ They make predictions about things **beyond** the data
- ▶ This requires some assumptions about the **data-generating process**.

Models

Models as summaries

- ▶ They summarise what we can see in the data
- ▶ The ultimate model of the data **is** the data

Models as predictors

- ▶ They make predictions about things **beyond** the data
- ▶ This requires some assumptions about the **data-generating process**.

Example models

- ▶ A numerical mean
- ▶ A linear classifier
- ▶ A linear regressor
- ▶ A deep neural network
- ▶ A Gaussian process
- ▶ A large language model

Estimates and decisions

We always need to make decisions based on some **estimates**.

Estimates and decisions

We always need to make decisions based on some **estimates**.

Estimate the bias of a coin

- ▶ I give you a coin that, lands with some fixed probability on heads.
- ▶ You are allowed to experiment with the coin.
- ▶ I will pay you **1 CHF** if you guess the throw correctly
- ▶ Otherwise you pay me **x CHF**.
- ▶ How much should I ask you to **pay** for the bet to be **fair**?
- ▶ What do you need to **know** to determine this?

Estimates and decisions

We always need to make decisions based on some **estimates**.

Estimate the bias of a coin

- ▶ I give you a coin that, lands with some fixed probability on heads.
- ▶ You are allowed to experiment with the coin.
- ▶ I will pay you **1 CHF** if you guess the throw correctly
- ▶ Otherwise you pay me **x CHF**.
- ▶ How much should I ask you to **pay** for the bet to be **fair**?
- ▶ What do you need to **know** to determine this?

Example (If the coin is fair)

- ▶ If the coin is fair, then you only have 50% probability of guessing correctly.
- ▶ If you bet x CHF, your expected return is x

The Bernoulli distribution

The Bernoulli distribution

Definition (Bernoulli distribution)

We say that $x \in \{0, 1\}$ has Bernoulli distribution with parameter θ and write

$$x \sim \text{Bernoulli}(\theta),$$

when

$$\mathbb{P}(x) = \begin{cases} \theta & x = 1 \\ 1 - \theta & x = 0. \end{cases}$$

The Bernoulli distribution

Definition (Bernoulli distribution)

We say that $x \in \{0, 1\}$ has Bernoulli distribution with parameter θ and write

$$x \sim \text{Bernoulli}(\theta),$$

when

$$\mathbb{P}(x) = \begin{cases} \theta & x = 1 \\ 1 - \theta & x = 0. \end{cases}$$

Example (Applications of the Bernoulli distribution)

- ▶ A biased coin flip.
- ▶ Classification errors.

The Bernoulli distribution

Definition (Bernoulli distribution)

We say that $x \in \{0, 1\}$ has Bernoulli distribution with parameter θ and write

$$x \sim \text{Bernoulli}(\theta),$$

when

$$\mathbb{P}(x) = \begin{cases} \theta & x = 1 \\ 1 - \theta & x = 0. \end{cases}$$

Example (Applications of the Bernoulli distribution)

- ▶ A biased coin flip.
- ▶ Classification errors.

Exercise: The expected value

If x is Bernoulli with parameter θ , then what is the expected value of

- ▶ The variable $f(x) = x - 1$?
- ▶ The variable $g(x) = (x - 1)^2$?

Two-variable models

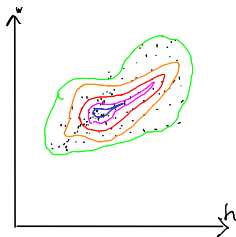


Figure: $x \in \mathbb{R}^2$

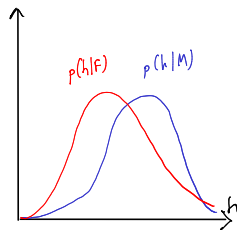
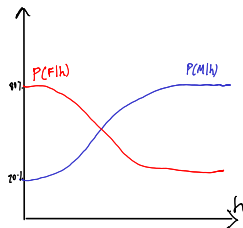
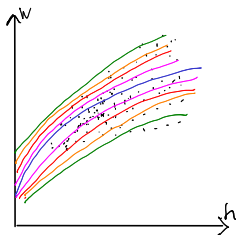


Figure: $x \in \mathbb{N} \rightarrow y \in \mathbb{R}$



Predicting y from x , discrete case.

Consider two variables, x, y . We can either care about

- ▶ $\mathbb{E}[y|x]$ the expectation of y for all x .
- ▶ $\mathbb{P}[y|x]$ the distribution of y for all x .

Probability table for $P(x, y)$

$P(x, y)$	$y = 0$	$y = 1$
$x = 0$	54%	6%
$x = 1$	16%	24%

- ▶ How can we graph this?
- ▶ What is $P(x)$?

Conditional probability table for $P(y|x)$

$P(y x)$	$y = 0$	$y = 1$
$x = 0$	90%	10%
$x = 1$	40%	60%

- ▶ What is $\mathbb{E}[y | x]$?

Distributions of two variables

In this setting, both x and y have a Bernoulli distribution. If we use a model whereby x is sampled first, and then y , then we can define two Bernoulli distributions. The first, for x is unconditional, while the second, for y , depends on the value of x :

$$\begin{aligned}x &\sim \text{Bernoulli}(\theta) \\ y \mid x &\sim \text{Bernoulli}(\phi_x).\end{aligned}$$

In our example, $\phi_0 = 0.1$ and $\phi_1 = 0.6$.

Homework

Probability table for $P(x, y)$

$P(x, y)$	$y = -1$	$y = 0$	$y = 1$
$x = 0$	10%	20%	10%
$x = 1$	30%	20%	10%

Given the above table, calculate

- ▶ $P(x)$
- ▶ The conditional probability table for $P(y|x)$.
- ▶ $\mathbb{E}_P[f]$ where $f(x, y) = x + y$.
- ▶ $\mathbb{E}_P[g]$ where $g(x, y) = xy$.
- ▶ $\mathbb{E}[y|x]$ for all values of x .

Two variables: conditional expectation

The height of different genders

The conditional expected height

$$\mathbb{E}[h \mid g = 1] = \sum_{\omega \in \Omega} h(\omega) P[\omega \mid g(\omega) = 1]$$

The empirical conditional expectation

$$\mathbb{E}[h \mid g = 1] \approx \frac{\sum_{t: g(\omega_t)=1} h(\omega_t)}{|\{t : g(\omega_t) = 1\}|}$$

Python implementation

Two variables: conditional expectation

The height of different genders

The conditional expected height

$$\mathbb{E}[h \mid g = 1] = \sum_{\omega \in \Omega} h(\omega) P[\omega \mid g(\omega) = 1]$$

The empirical conditional expectation

$$\mathbb{E}[h \mid g = 1] \approx \frac{\sum_{t: g(\omega_t)=1} h(\omega_t)}{|\{t : g(\omega_t) = 1\}|}$$

Python implementation

```
h[g==1] / sum(g==1)
## alternative
import numpy as np
np.mean(h[g==1])
```

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Pandas and dataframes

Additional notes on pandas

Single variable models

Two variable models

Statistics, validation and model selection

Course summary

Course Contents

For next week

Assignment

Reading

Populations, samples, and distributions

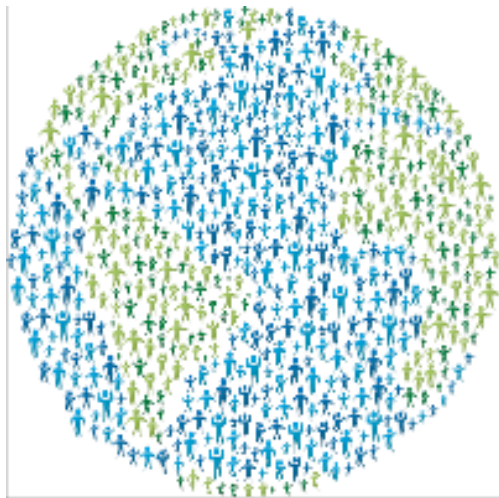


Figure: The world population



Figure: A sample

Statistical assumptions

Independent, Identically Distributed data

- ▶ $\omega_t \sim P$: individuals $\omega_t \in \Omega$ are drawn from some **distribution** P
- ▶ $\mathbf{x}_t \triangleq \mathbf{x}(\omega_t)$ are some **features** of the t -th individual
- ▶ Here we are interested in properties of the **unknown** distribution P .

Representative sample from a fixed population

- ▶ Finite population $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$
- ▶ A subset $S \subset \Omega$ of size $T < N$ is selected with a **uniform distribution**, i.e. so that

$$P(S) = T/N, \quad \forall S \subset \Omega.$$

- ▶ Here we are interested in statistics of the **unknown** population Ω .
- ▶ We assume an underlying distribution P for convenience.
- ▶ We can try both cases essentially the same.

Learning from data

Unsupervised learning

- ▶ Given data $x_1, \dots, x_T$.
- ▶ Learn about the data-generating process.
- ▶ Example: Estimation, compression, text/image generation

Learning from data

Unsupervised learning

- ▶ Given data $x_1, \dots, x_T$.
- ▶ Learn about the data-generating process.
- ▶ Example: Estimation, compression, text/image generation

Supervised learning

- ▶ Given data $(x_1, y_1), \dots, (x_T, y_T)$
- ▶ Learn about the relationship between x_t and y_t .
- ▶ Example: Classification, Regression

Learning from data

Unsupervised learning

- ▶ Given data $x_1, \dots, x_T$.
- ▶ Learn about the data-generating process.
- ▶ Example: Estimation, compression, text/image generation

Supervised learning

- ▶ Given data $(x_1, y_1), \dots, (x_T, y_T)$
- ▶ Learn about the relationship between x_t and y_t .
- ▶ Example: Classification, Regression

Online learning

- ▶ Sequence prediction: At each step t , predict x_{t+1} from $x_1, \dots, x_t$.
- ▶ Conditional prediction: At each step t , predict y_{t+1} from $x_1, y_1, \dots, x_t, y_t, x_{t+1}$

Learning from data

Unsupervised learning

- ▶ Given data $x_1, \dots, x_T$.
- ▶ Learn about the data-generating process.
- ▶ Example: Estimation, compression, text/image generation

Supervised learning

- ▶ Given data $(x_1, y_1), \dots, (x_T, y_T)$
- ▶ Learn about the relationship between x_t and y_t .
- ▶ Example: Classification, Regression

Online learning

- ▶ Sequence prediction: At each step t , predict x_{t+1} from $x_1, \dots, x_t$.
- ▶ Conditional prediction: At each step t , predict y_{t+1} from $x_1, y_1, \dots, x_t, y_t, x_{t+1}$

Reinforcement learning

Learn to act in an **unknown** world through interaction and rewards

Data, models, and reproducibility.

Training data

- ▶ Calculations, optimisation
- ▶ Data exploration

Data, models, and reproducibility.

Training data

- ▶ Calculations, optimisation
- ▶ Data exploration

Validation data

- ▶ Fine-tuning
- ▶ Model selection

Data, models, and reproducibility.

Training data

- ▶ Calculations, optimisation
- ▶ Data exploration

Validation data

- ▶ Fine-tuning
- ▶ Model selection

Test data

- ▶ Performance comparison

Data, models, and reproducibility.

Training data

- ▶ Calculations, optimisation
- ▶ Data exploration

Validation data

- ▶ Fine-tuning
- ▶ Model selection

Test data

- ▶ Performance comparison

Simulation

- ▶ Interactive performance comparison
- ▶ White box testing

Data, models, and reproducibility.

Training data

- ▶ Calculations, optimisation
- ▶ Data exploration

Validation data

- ▶ Fine-tuning
- ▶ Model selection

Test data

- ▶ Performance comparison

Simulation

- ▶ Interactive performance comparison
- ▶ White box testing

Real-world testing

- ▶ Actual performance measurement

Model selection

- ▶ Train/Test/Validate
- ▶ Cross-validation
- ▶ Simulation

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Pandas and dataframes

Additional notes on pandas

Single variable models

Two variable models

Statistics, validation and model selection

Course summary

Course Contents

For next week

Assignment

Reading

Course Contents

Models

- ▶ k-Nearest Neighbours.
- ▶ Linear models and perceptrons.
- ▶ Multi-layer perceptrons (aka deep neural networks).
- ▶ Bayesian Networks

Algorithms

- ▶ (Stochastic) Gradient Descent.
- ▶ Bayesian inference.

Reproducibility

- ▶ Modelling assumptions
- ▶ Interactions and feedback

Fairness

- ▶ Implicit biases in training data
- ▶ Fair decision rules and meritocracy

The problems of Machine Learning (1 week)

Introduction

Estimation

Answering a scientific problem

Pandas and dataframes

Additional notes on pandas

Single variable models

Two variable models

Statistics, validation and model selection

Course summary

Course Contents

For next week

Assignment

Reading

Assignment

► See Moodle

Reading for this week

ISLP Chapter 1